RHSOi

River Hill Science Olympiad Invitational 2021

Water Quality



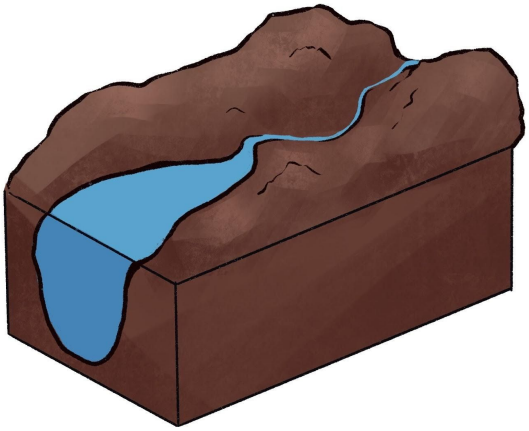
Answer Key

Total Points: _____/156.03

Victoria Cheng/vcheng2003@gmail.com

Kianna Pan/kiannapan@gmail.com

Marine and Estuary Ecology: 50 points



1. Identify the type of estuary (1 point)

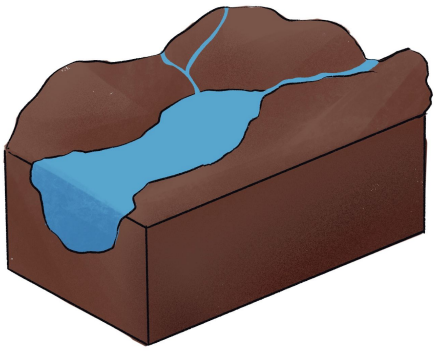
Fjord

2. Explain how this type of estuary is formed (2 points)

Created by glaciers. Fjord estuaries occur when glaciers carve out a deep, steep valley. Glaciers retreat and the ocean rushes into the narrow, deep depression.

3. Provide a real world example of this type of estuary: (1 point)

Fjords are also found throughout Canada, Chile, New Zealand, Greenland, Norway, Siberia, and Scotland. Puget Sound is a series of fjord estuaries in the U.S. state of Washington.



4. Identify the type of estuary (1 point)

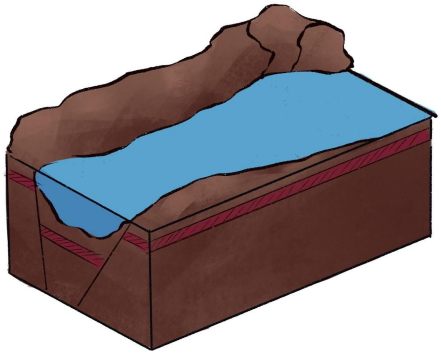
Coastal plain/ drowned river mouth

5. Explain how this type of estuary is formed (2 points)

Sea level rise and flooded low lying river valleys

6. Provide a real world example of this type of estuary (1 point)

The Chesapeake Bay on the East Coast of the United States and Coos Estuary on the West Coast

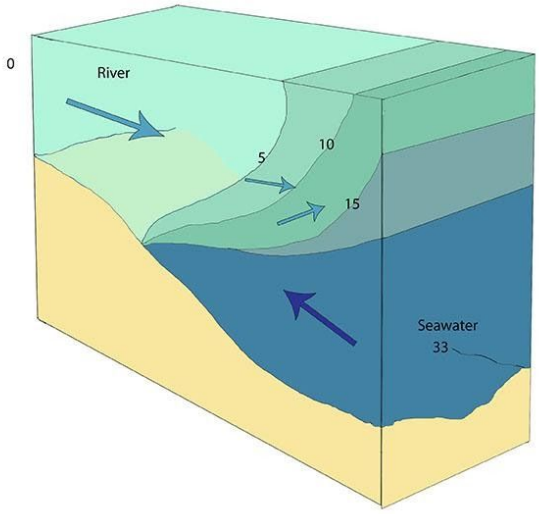


7. Identify the type of estuary (1 point)

Tectonic or Fault Bay

8. Where are these types of estuaries found? (1 point)

(along fault lines such as Manakau Harbour), California's San Francisco Bay



9. Classify this estuary based on salinity and water circulation (1 point)

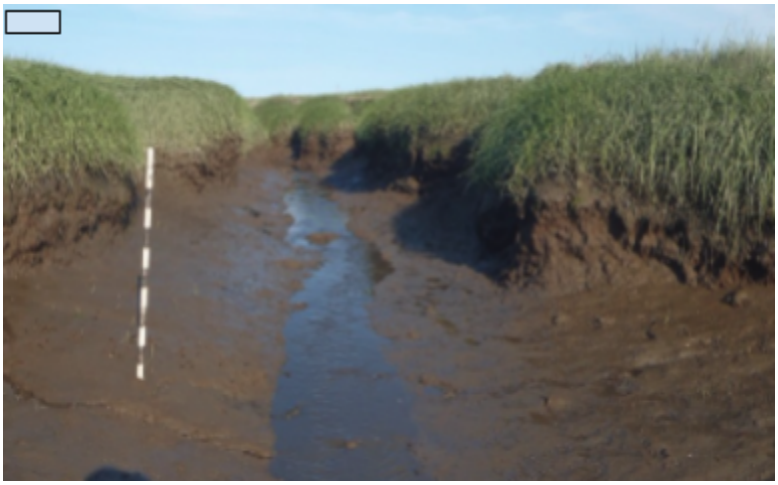
Salt wedge, highly stratified estuaries

10. Explain the specific difference between a well mixed and partially mixed estuary (3 points)

Partially: As tidal forcing increases, river output becomes less than the seawater input.
Well mixed: Tidal mixing forces exceed river output, resulting in a well-mixed water column.
" similar salinity profiles to vertically mixed estuaries, where salinity increases from the head to the mouth, but there is also a slight increase in salinity with depth at any point."

11. Where do the majority of pollutants that enter estuaries come from? (1 point)

a. Land based sources
b. Marine dumping





12. Explain why the second creek does not look like the first, and how this issue negatively impacts estuaries. (4 points)

The first creek has normal nutrient levels, while the second creek has excessive nutrient pollution. "The belowground structure (biomass) of salt marsh plants (the root system) is responsible for nutrient uptake and storage. The roots and rhizomes of cordgrass (Spartina alterniflora) are key support structures for the salt marsh because they hold together the peat, clay and silt composition of the marsh. When estuaries are flooded with nutrient-polluted water, cordgrass receives ample access to nutrients and does not need to grow extensive root systems. Consequently, the belowground biomass of cordgrass is smaller, leading to weaker, less stable creek walls. This response ultimately contributes to the cracking, collapsing, and decomposing into unvegetated mud events that are documented in nutrient-polluted estuaries."
Effect on estuaries- eutrophication/algae, primary production, coastal resilience, habitat, etc

13. Victoria is incredibly passionate about her local estuary and has been noticing its decline in health. She wants to propose ideas to local government members, teachers, farmers, and business owners. What ideas can she advocate for to combat this issue in her community? (2 points)

Reduce fertilizer by improving soil quality, green spaces, rain gardens, permeable hard surfaces, choosing compost, limit use of phosphate. Till planting

14. How does sedimentation affect estuary organisms? (2 points)

Increases Turbidity, so aquatic plants can photosynthesize to make food, predators can't find prey, smothers the seabed and kills mud-dwellers that can't burrow up quickly can sometimes also clog fish gills

15. Which of the following is caused by estuary water movement?

a. Transport organisms
b. Increased dissolved oxygen
c. Protection of the organisms
d. Circulation of nutrients and oxygen
e. Transport sediments and wastes

16. Describe how water bugs are able to stand on water. (1 point)

Surface tension due to cohesion of water molecules. Bugs also have hydrophobic hairs on their feet.

17. What are some features that may vary along the length of an estuary? (1 point)

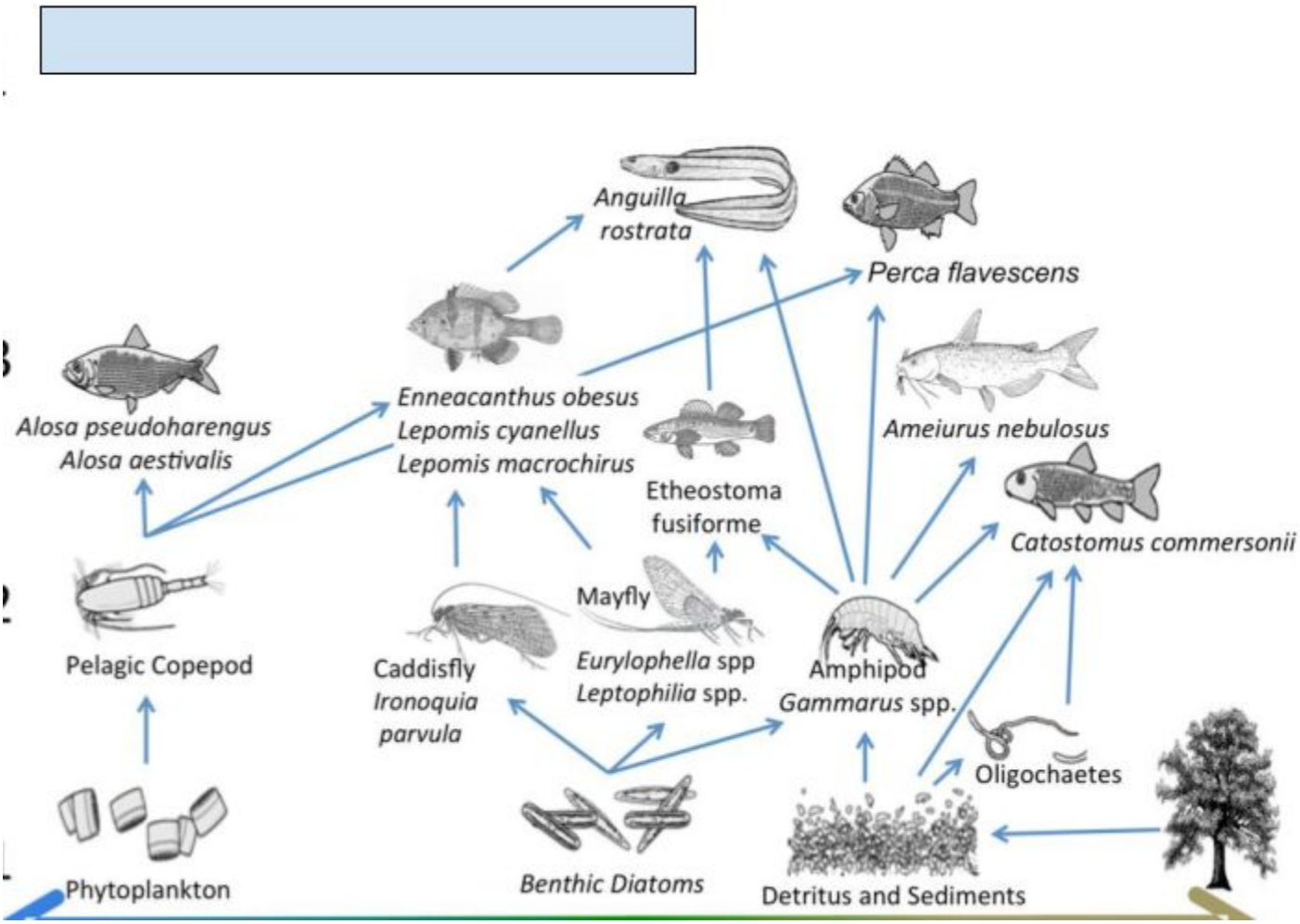
Examples: Temperature, Light levels, Salinity

18. What is the bottleneck effect? (2 points)

A genetic drift that occurs when part of a population is suddenly reduced due to natural disasters.

19. Which of the following can cause a bottleneck effect:

a. **Volcano eruption**
b. **Increase in salinity**
c. **Over-hunting**
d. **Deforestation**
e. **Geographic isolation**
f. **interbreeding**



20. Give an example of a food chain in the food web (2 points)

Detritus, amphipod, anguilla rostrata

21. How are mangroves maintain water quality in an estuary? Order the types mangroves from closest to farthest from the ocean: Black mangroves, red mangroves, white mangroves. (2 points)

They can prevent erosions by stabilizing sediments with their roots, roots can filter and trap pollutants, protect from natural disasters, absorb nutrients from runoff. Red, black, white

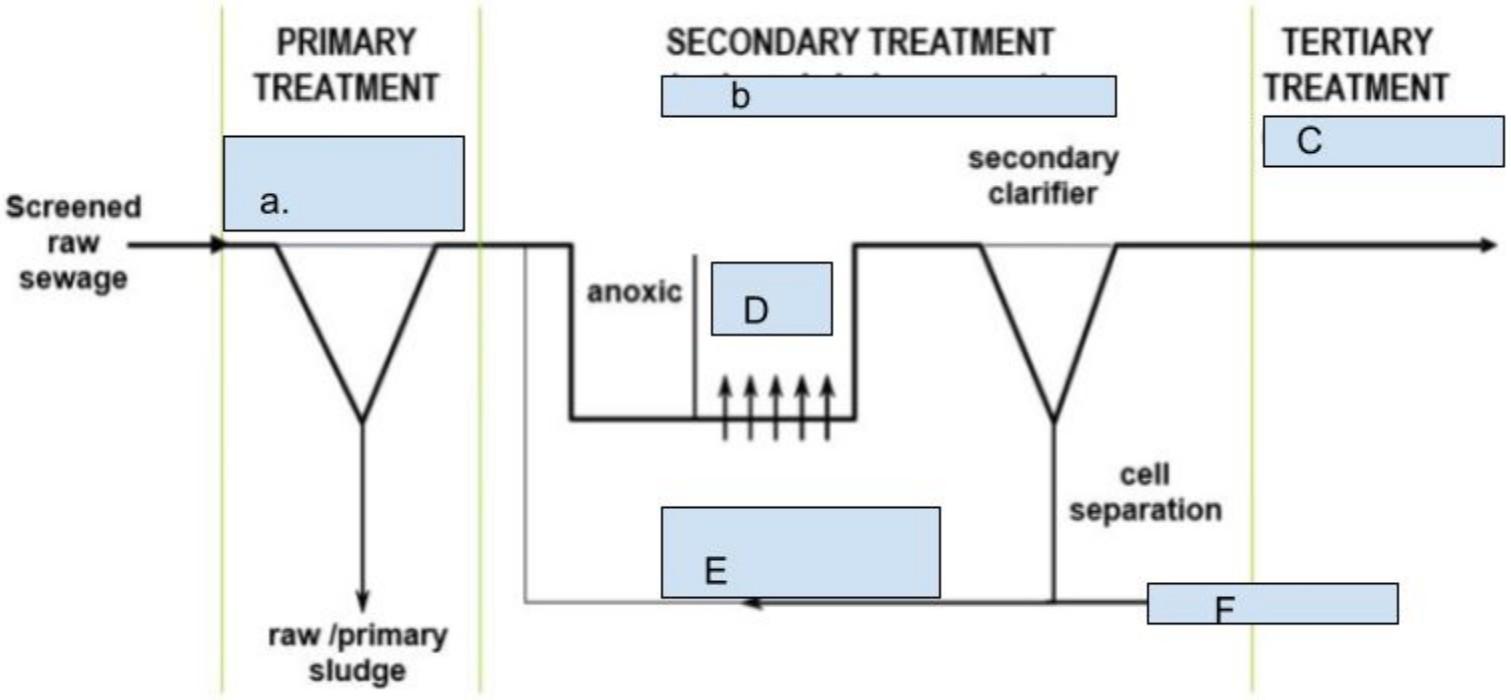
22. Who is the apex predator in the food web? (1 point)

Alosa pseudoharengus, anguilla rostrata, perca flavescens

23. Order the steps of Potable water treatment: Filtration, coagulation and flocculation, disinfection, sedimentation. (2 points)

Coagulation and flocculation: mixing of positively charges particles so they will clump with negatively charged suspended solids. Flocculation increases particle size. Flocs will settle when they're large. It will be mixed gently to preserve flocs
Sedimentation: rapid mixing to disperse coagulents and particle collisions
Filtration: Use inorganic coagulants
Disinfection: polymers are added which will be effective over a wide range of pHs and preserve alkalinity

24. Label the missing steps and describe what should be happening in those steps of wastewater treatment.(6 points)



A. Primary sedimentation/sedimentation
B. Activated sludge process
C. Secondary clarifier
D. Aerobic
E. Recycle for denitrification
F. Waste sludge

25. Name 2 downsides to a septic tank. (2 points)

Periodic maintenance, ruptured pipes, easily clogged (especially if u flush things you're not supposed to)

26. How could a broken septic tank affect the nearby flora and fauna of a body of water? (1 point)

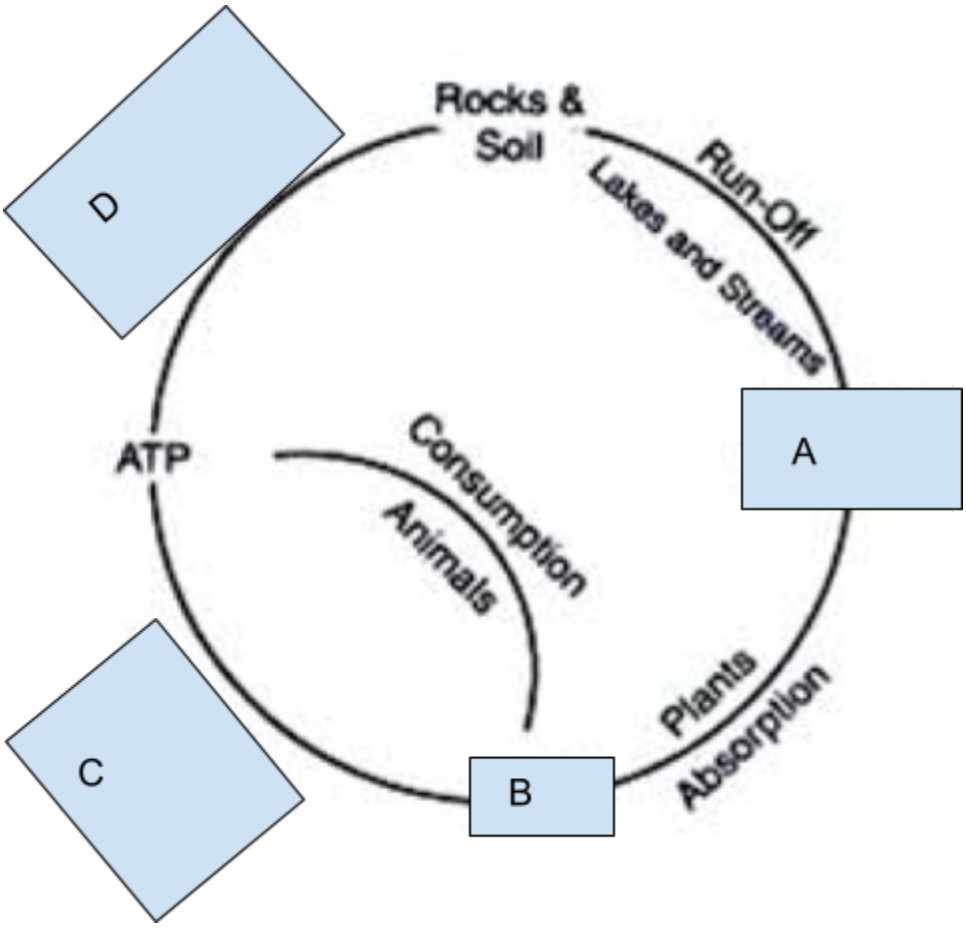
Increase fecal matter in nearby streams, waste could fertilize the algae and which decreases dissolved oxygen and limits sunlight

27. What is the difference between a r selected and k selected species? (2 points)

Note: R and K selected organisms are is a different classification from Type I, II, III. Type I, II, III survivorship curves refer to the survival rates of classified organisms.
R selected species are known as “boom” and “bust” they are larger and use a lot of energy. They tend to have less offspring and a longer life expectancy. They reproduce more than once in their lifetime and are density dependent.
K selected species are smaller and use less energy. They have a shorter life expectancy and usually only reproduce once in their life time and are density independent.

28. Name two examples of r selected species (1 point)

Plants, froggies, fishies, mice



29. Fill in the missing blanks of the phosphorus cycle. (2 points)

A. Inorganic phosphate
B. PO4 (3-)
C. Metabolism
D. Decompisiont

Coral Reef Microflora and fauna: 50 points



30. Identify the specimen (1 point)

Sweetlips

31. Is this a juvenile or an adult? (1 point)

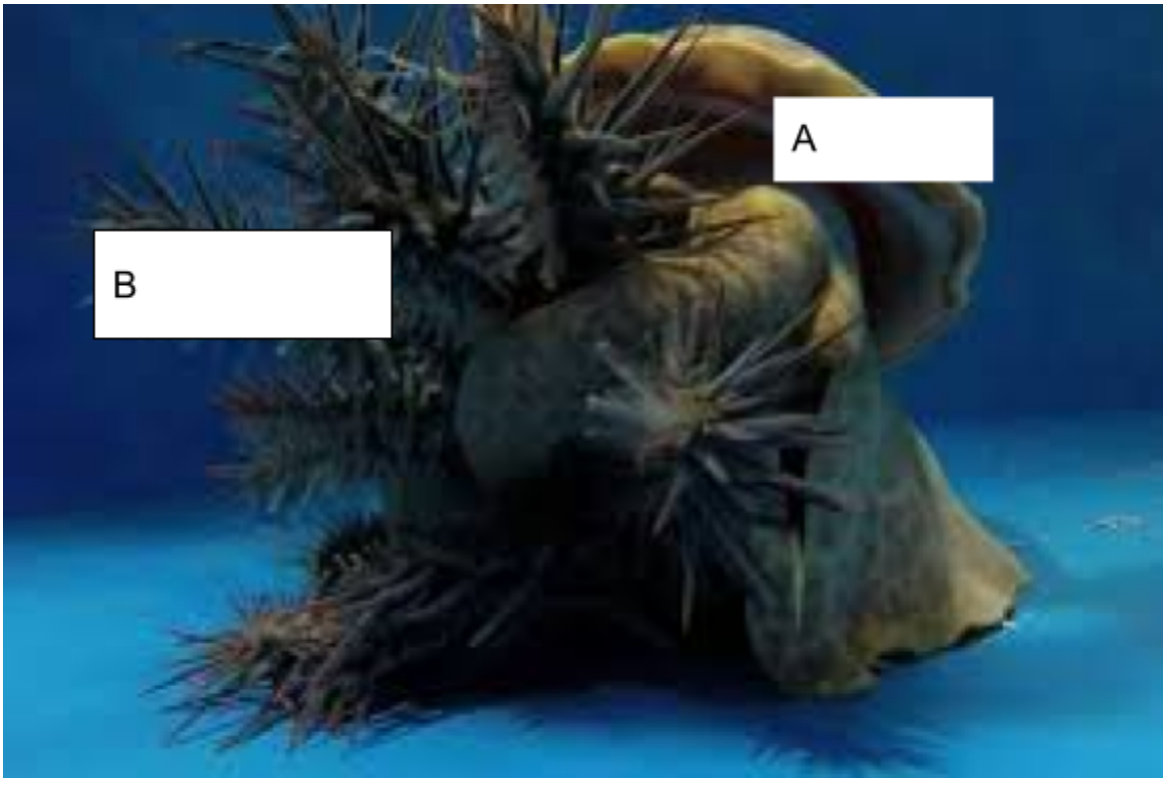
Juvenile

32. What is one adaptation these organisms have developed to help themselves eat. (1 point)

Nocturnal, colonies, etc.

33. What other organism on this year's list has the same diet? (1 point)

Red snapper, humphead wrasse



34. Identify organism A (1 point)

Triton

35. Identify organism B (1 point)

Crown of Thorns Starfish

36. What relationship do these organisms have? (1 point)

Predator and Prey

37. Where does specimen B get its name from? (1 point)

Spines resemble biblical crown of thorns

38. If a human were to touch specimen B, what should they do? (1 point)

They do not inject toxins, but the tissues with the aterosaponins are left in the wound. Sharp stinging pain with bleeding. Surgically remove spines

39. What is the importance of specimen A to coral reefs? (1 point)

Specimen A is a keystone species and is a good indicator of coral health. They also eat crown of thorns starfish.

40. What happens during coral asexual reproduction? (2 points)

Budding: little polyps bud off to grow on their own
Parthenogenesis: using fragments of corals; embryos growing without fertilization
Coral bail out: cloning; single coral polyp splits from the adult and drifts somewhere else to develop

41. What is spawning? (1 point)

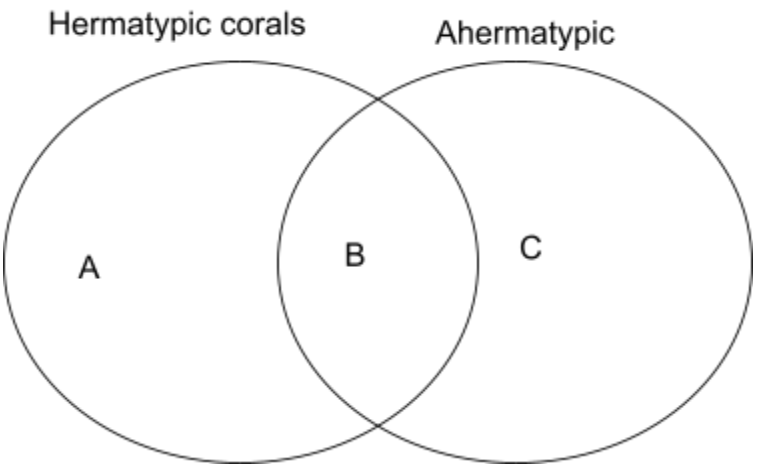
Corals release eggs and sperm into the water column. The sperm fertilizes the eggs and the larvae are released into water where they develop.

42. There is currently a new moon and the temperature of the water is around 25 degrees celsius. Will coral spawning occur? (1 point)

no

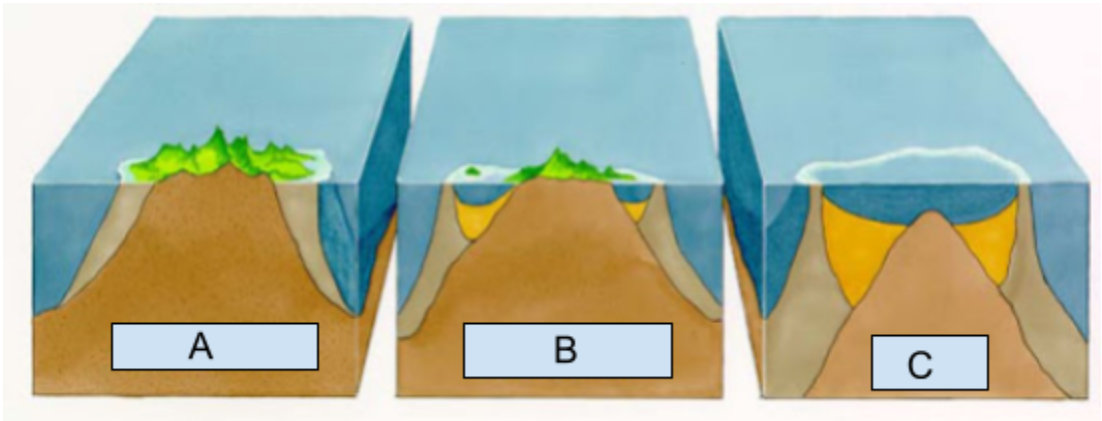
43. Place the following phrases in A, B, or C (3.5 points)

- Hard coral
- Soft coral
- Do not require zooxanthellae
- Reef building
- Calcareous skeleton
- Deep water reefs
- Have colonies



A: 1457
B: 6
C: 23

44. Label the following diagram of coral reef formation. (1.5 points)



A: Fringing B: Barrier C: Atoll

45. Which of the following is required for coral reef formation? (2 points)

A. Low nutrient waters
B. Solid base
C. High temperatures
D. High salinity
E. Water circulation
F. All of the above

46. What is biotic potential? What are the factors that affect biotic potential? (2 points)

The maximum ability for an organism to reproduce. Four factors: space, light, substrate, chemical compounds, predators and parasites

47. Why do most species never reach biotic potential and why could biotic potential be bad? (2 points)

Nature isn't perfect and species rarely live under "ideal environmental conditions." If a population were to reach biotic potential then there's a high possibility they will become invasive and negatively affect the flora and fauna around them.

48. What is zooxanthellae? What relationship do they have with coral reefs? (2 points)

Zooxanthellae are bacteria. Theyl have a symbiotic relationship with coral. Zooxanthellae photosynthesize for the coral and the coral offers them protection.

49. How can overfertilizing farms negatively affect coral reefs? (2 points)

Overfertilizing farms leads to eutrophication which blocks the sunlight from the zooxanthellae which means that the Eutrophication also decreases Dissolved Oxygen.

50. In the past few years, there has been a drastic decrease of butterfly fish. What is the reason for their decrease and what has been done to bring back their populations? (2 points)

Butterfly fish rely on coral reefs to reproduce and feed on tiny crustaceans and planktonic animals found in the crevices of corals. Since coral reefs have been on a decline due to the rising temperatures, butterfly fish have lost their habitat. Scientists have tried to restore their habitats.

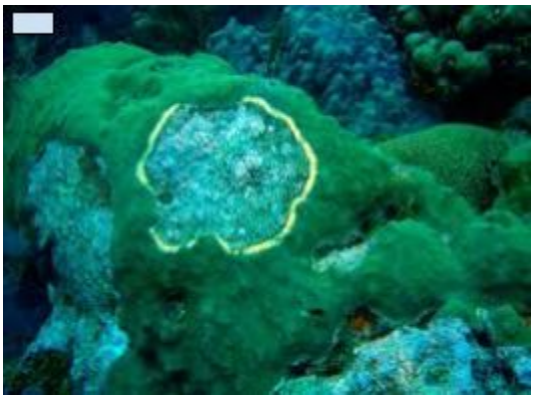


51. What disease is displayed? (1 point)

Black band disease

52. How does it affect the coral? (1 point)

Destroys live tissue and only leaves behind the skeleton. It can kill colonies due to lack of oxygen.



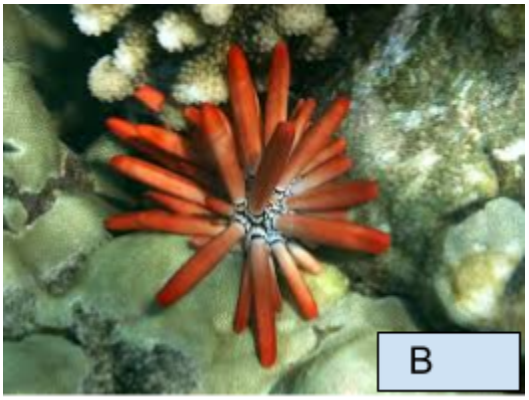
53. What is disease is displayed? (1 point)

Yellow band disease

54. How does it affect the coral? (1 point)

Kill zooxanthellae, coral loses a lot of energy, coral starves to death





55. Identify Specimen A (1 point)

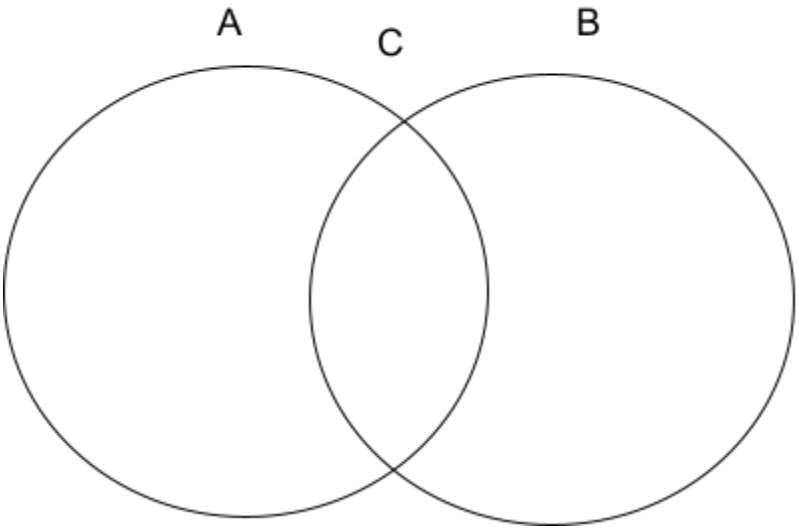
Long-spined black sea urchin

56. Identify Specimen B (1 point)

Pencil Urchin

57. Place the following terms in A, B, or C (4 points)

- Prevent algae overgrowth
- Shallow water sea urchins
- Atlantic Ocean, Carribean Ocean, Gulf of Mexico,
- Move 96 cm per day
- Magnesium calcite spines
- Tropical urchins
- Had a major “die off” in the 1980s
- 30 cm long sea urchins



A: 2378 B: 456 C: 1





58. Identify Specimen A (1 point)

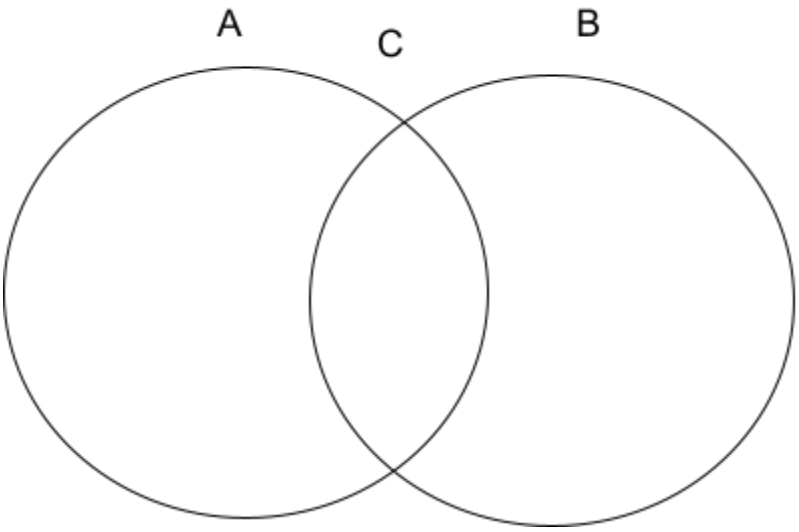
Humphead Wrasse

59. Identify Specimen B (1 point)

Bumphead Parrotfish

60. Place the following terms in A, B, or C (4 points)

- Indian and Pacific Oceans
- Vertical head profile
- Uniformly covered in scales
- Sexually monochromatic
- Spawns according to the lunar cycle
- Protogynous hermaphrodites
- Endangered on the IUCN Red List
- Cooperatively hunt with groupers



A: 2345
B: 6
C: 178

61. Explain what happens during coral bleaching? (2 points)

Coral bleaching typically occurs there when the water is too warm or has too much sediment or overfishing. They lose their vibrant colors because they will expel zooxanthellae. It destroys ecosystems because animals that depend on coral reefs for food and shelter must look elsewhere.

62. Which of the following are reasons coral bleaching may occur? (2 points)

A. Sunscreen
B. Rising temperatures
C. Too much sunlight
D. High tides
E. All of the above

63. What is the El Niño? (1 point)

Periodic warming in sea surface temperatures across the central and east-central Equatorial Pacific. It reduces the upwelling of cold water off the coast of the Americas.

64. How does it correlate with coral bleaching? (1 point)

El Niño raises ocean temperatures. The sudden change in temperature can cause corals to expel zooxanthellae. Even though El Niño is short term, its effects on coral are long term.

65. What is an example of interference competition in Coral Reefs? (1 point)

Interference competition: two species compete over the same scarce resource. Snapper and bumphead parrotfish

Water Monitoring and Analysis: 50 points

Bottle Number	mL Seed	Initial DO	Final DO	Depletion
1	2	5.9	2.8	3.1

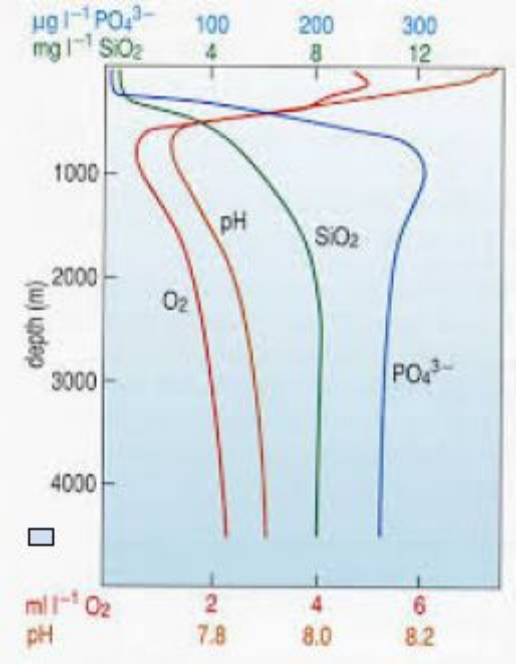
2	4	5.8	2.0	3.0
3	8	5.9	1.5	4.4
4	12	5.3	2.7	2.9

66. Kianna is running around the beach, chasing birds and comes across some interesting seeds! Like the environmental science nerd she is, she's interested in the BOD of the seeds on the beach. What is the BOD of each seed dilution and what is the average seed BOD? (5 points)

Bottle 1: 465
Bottle 2: 285
Bottle 3: 165
Bottle 4: 260
Average: 293.75

67. What does the average seed BOD that you calculated tell you about the sample and the water it comes from? (2 points)

293.75 mg/L BOD is exerted by seed material. A 300 mL of undiluted seed material should be 293.75 mg/L DO. The water is heavily untreated.



68. What is the relationship between oxygen and pH? Explain why this might be the case from a chemistry point of view. (4 points)

O2 increases as pH increases. More oxygen means decrease in CO2 and therefore oxygen is added, they bond with H+ ions to from OH, which then increases the pH as it reduces the concentration of H+ ions and increases the hydroxyl.

69. Jeff goes to talk to you, the environmental teacher. He is wondering what the aragonite saturation before organisms become stressed. What do you tell him? (1 point)

3 omega

70. Can you tell Jeff what the balanced chemical equation for aragonite is? (1 point)

Ca + CO3 ⟷ CaCO3

71. Jeff proceeds to do some tests on the creek. He plans on testing pH, what is the best tool to use for testing pH? (2 points)

pH strips, litmus paper, pH probs, etc.

72. What could increase the pH of a stream? (1 point)

Acid rain, lime stone, salinity, whatever creative answers people came up with.

73. If the average pH of a stream is 7 and the average pH of the ocean is 8.0, how many times more acidic is the ocean? Show work below. (2 points)

I messed up the wording of this question so if you put 1/10 or 10 I gave you a point and work:

74. As temperature increases, what happens to dissolved oxygen? (1 point)

Decreases

75. What units are used to measure dissolved oxygen? (1 point)

mg/L, ppm, umol

76. What effect does overgrown algae have on dissolved oxygen? (2 points)

Overgrown algae decreases dissolves oxygen. Even though they are producers, when they die, the amount of oxygen used for decomposing is more than the oxygen produced.

78. What amount of ppm of Dissolved oxygen can put aquatic life under stress? What amount of ppm of Dissolved oxygen can put large fish under stress? (2 points)

Below 5.0 mg/L - aquatic life
Below 1-2 mg/L - large fish

79. Which of the following will happen when there's an increased salinity? (2 points)

a. Conductivity decreases
b. Seawater becomes denser
c. Dissolved Oxygen increases
d. None of the above

80. Define the following terms: homoiohaline, poikilohaline, hyperhaline, metahaline, polyhaline, mesohaline, and oligohaline. (0.5 points each, 3.5 points total).

homoio: salinity does not vary (constant)

poikilo: 0.5-300ppt

hyper: 80-300ppt

meta: 40-60ppt

poly: 18-30ppt

meso: 5-18ppt

oligo: 0.5-5ppt

81. What is an estuary? (1 point)

an area of water where like a source of freshwater and the salty ocean merge

82. To celebrate the start RHSOI, Saketh decided to set off some fireworks over the water. However, since Saketh isn't too aware of the consequences of fireworks, how could Saketh's celebrations negatively affect the water's turbidity? What are some effects of the turbidity? (3 points)

Turbidity can cloud the water and block photosynthesis. The sediments from Saketh's fireworks can get into the fish's gills. It can also make it hard for fish to see through the water.

83. One of the main components of fireworks is sulfur. How could sulfur affect the drinking water and piping in the area? (2 points)

The sulfur can damage piping made of copper and drinking water with sulfur can cause diarrhea and dehydration.

84. Would this be considered a point source or non point source? (1 point)

Point source

85. River Hill is surrounded by many farms. There are cow farms, corn mazes, pumpkin patches and more! There are also many small creeks that lead to the Middle Patuxent River. What are some consequences the farms could have on the Patuxent River? Are the farms considered a point or nonpoint source? (3 points)

Farms are a nonpoint source. The fertilizers could increases algae blooms which affects DO and the flora and fauna in the river. The pesticides could contaminate drinking water, sediments from erosion, manure could increase fecal coliform.

86. Name two things that could increase fecal coliform in a body of water. (2 points)

Manure from farms, broken septic systems, sewage

87. Name four waterborne pathogens that could come from fecal coliform? (0.5 each, 2 points)

Dysentery, typhoid, gastroenteritis, hepatitis A, ear infections

88. How is water hardness measured?

Grains per gallon, milligrams of calcium per liter, ppm, dH

Scenario: Red Tide

The Red Tide is caused by algal blooms when algae blooms are so big that they discolor coastal weathers. It happens once a year in the Gulf of Mexico and is known to occur once a year.

89. What are the harmful effects of the algal blooms on the water environment

Toxins, dead zones, block sunlight, decrease DO

90. What are the harmful effects of the algal blooms to humans?

Can produce toxins that can make fish and shellfish dangerous to eat. They also make the air difficult to breathe and no one can swim in the water.

91. What type of water can they be found in?

Red Tide is found in the Gulf of Mexico. Red tide, in particular, occur in warm water, particularly oceans, bays, and estuaries.

These algal blooms are known to have natural toxins.

92. What is HAB?

Harmful Algae Blooms

93. Define biomagnification and bioaccumulation

Biomagnification: Accumulation of toxins in organisms
Bioaccumulation: Net accumulation of toxins over time

94. How could biomagnification/bioaccumulation of algal be detrimental to larger organisms and the humans in the area?

Since the algae has toxins, the animals that feed on the algae will accumulate the toxins and the humans that later consume the fish in the area will consume these toxins.

95. Large fish are immune to the toxins, however they still eat the algae. What could this mean to humans that consume the fish?

Humans that consume the fish will consume the toxins.

96. How much does a person pay for water on a daily basis? (Tiebreaker)

0.25

97. How much of the world's water is frozen? (Tiebreaker)

2%